

DDoS Incident Report

Report Period	Sep 15, 2026 - Sep 15, 2026
Generated At	2026-09-15 19:53:06
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

Metric	Value	Traffic Summary	Count
Total Threats Mitigated	23	Total Flows Observed	179283
High Priority	13	Total Packets Analyzed	1330122522
Low Priority	10	Malicious Packets	1329382782
Manual Releases	0	True Negatives Passed	135792
Manual Blocks	0		
		Attack Vector	Count
		ICMP Flood	0
		SYN Flood	0
		UDP Flood	16
		Uncertain	7
ML Metrics	Value	Action	Count
False Positives	45	Quarantined	4
FP Rate	0.02%	Time Ban	3
IF Precision	99.95%	Blackhole	0
IF Recall	94.14%	Released	7
IF F1-Score	96.96%		
IF Accuracy	98.48%		

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.95%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.14%	Share of actual attacks successfully flagged
F1-Score	96.96%	Balanced measure combining Precision and Recall
Accuracy	98.48%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.02%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.86%	Actual attacks that went undetected
True Positive Rate (TPR)	94.14%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.98%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 43343	FN 2696
Actual: Normal	FP 21	TN 133096

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	43343

2c. Response Latency

Metric	Value	Description
Detection Time	264.2 ms	Average time taken to detect an attack
Mitigation Response Time	3.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.00%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	1.77%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.70%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.20	2	2	2026-09-15 19:50:30	2026-09-15 19:51:40
10.0.0.21	2	2	2026-09-15 19:50:30	2026-09-15 19:51:40
10.0.0.22	2	2	2026-09-15 19:50:33	2026-09-15 19:51:43

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 10:34:56	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:34:56	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:34:57	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 10:35:06	10.0.0.20	DDoS	UDP Flood	75.9%	High	Time Ban (1m)
2026-09-15 10:35:06	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-15 10:35:07	10.0.0.20	DDoS	UDP Flood	83.3%	High	Time Ban
2026-09-15 10:35:07	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-15 10:35:07	10.0.0.22	DDoS	UDP Flood	86.4%	High	Time Ban (1m)
2026-09-15 10:35:08	10.0.0.22	DDoS	UDP Flood	86.4%	Critical	Time Ban
2026-09-15 10:36:07	10.0.0.20	DDoS	UDP Flood	86.6%	High	Released
2026-09-15 10:36:07	10.0.0.21	DDoS	UDP Flood	79.1%	High	Released
2026-09-15 10:36:08	10.0.0.22	DDoS	UDP Flood	86.4%	High	Released
2026-09-15 10:36:18	10.0.0.20	DDoS	UDP Flood	86.6%	Critical	Time Ban (2m)
2026-09-15 10:36:19	10.0.0.22	DDoS	UDP Flood	86.4%	Critical	Time Ban (2m)
2026-09-15 10:41:51	10.0.0.6	DDoS	Uncertain	48.2%	Low	Released
2026-09-15 10:41:59	10.0.0.21	DDoS	UDP Flood	83.2%	High	Time Ban (2m)
2026-09-15 10:51:59	10.0.0.2	DDoS	Uncertain	61.1%	Low	Released
2026-09-15 11:10:35	10.0.0.10	DDoS	Uncertain	48.4%	Low	Released
2026-09-15 11:10:36	10.0.0.8	DDoS	Uncertain	64.1%	Low	Released
2026-09-15 11:10:45	10.0.0.9	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-15 11:12:07	10.0.0.8	DDoS	UDP Flood	90.9%	High	Time Ban (1m)
2026-09-15 11:22:45	10.0.0.6	DDoS	UDP Flood	85.1%	High	Time Ban (1m)
2026-09-15 11:24:20	10.0.0.6	DDoS	UDP Flood	92.4%	High	Quarantined

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6358	85.5%	Critical	Phase 2	2	1m 10s	2026-09-15 19:51:43	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6282	77.6%	High	Phase 2	2	1m 10s	2026-09-15 19:51:40	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6162	86.2%	High	Phase 2	2	1m 10s	2026-09-15 19:51:40	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6368	78.0%	High	Phase 2	1	10s	2026-09-15 19:50:43	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6330	75.6%	High	Phase 2	1	10s	2026-09-15 19:50:40	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6449	86.2%	High	Phase 2	1	10s	2026-09-15 19:50:40	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE